Example 8.17

In the experiment of rolling a pair of dice, the events A, B and C are defined as A : getting 2 on the first die, B : getting 2 on the second die, and C : sum of the faces of dice is an even number. Prove that the events are pair wise independent but not mutually independent?

Solution:

$$S = \{ (1,1), (1,2), (1,3), (1,4), (1,5), (1,6),$$
$$(2,1), (2,2), (2,3), (2,4), (2,5), (2,6),$$
$$(3,1), (3,2), (3,3), (3,4), (3,5), (3,6),$$
$$(4,1), (4,2), (4,3), (4,4), (4,5), (4,6),$$
$$(5,1), (5,2), (5,3), (5,4), (5,5), (5,6),$$
$$(6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \}$$
$$n(S) = 36$$

The outcomes which are favourable to the occurrence of these events can be listed below:

$$A = \{ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \}$$
$$n(A) = 6$$
$$P(A) = \frac{6}{36} = \frac{1}{6}$$
$$B = \{ (1,2), (2,2), (3,2), (4,2), (5,2), (6,2) \}$$
$$n(B) = 6$$
$$P(B) = \frac{6}{36} = \frac{1}{6}$$
$$C = \{ (1,1), (1,3), (1,5), (2,2), (2,4), (2,6), (3,1), (3,3), (3,5),$$
$$(4,2), (4,4), (4,6), (5,1), (5,3), (5,5), (6,2), (6,4), (6,6) \}$$
$$n(C) = 18$$
$$P(C) = \frac{18}{36} = \frac{1}{2} .$$
$$A \cap B = \{ (2,2) \}$$
$$n(A \cap B) = 1$$
$$P(A \cap B) = \frac{1}{36}$$